

The ML Lifecycle & CI/CD for ML

Google's "Hidden Technical Debt" paper opened every MLOps conversation. The other 95% — data, features, serving, monitoring — is where projects live or die. That 95% is what MLOps covers.

When did you last write data-validation tests alongside your model code?

L1

WHAT'S INSIDE

- 01** The 7-stage ML lifecycle
- 02** DevOps vs MLOps
- 03** 5 artefacts that must be versioned
- 04** CI/CD for ML — the pipeline
- 05** Analogy

WHY THIS LESSON MATTERS

ML code is less than 5% of a production ML system.

NUMBER 1

95%

From real-world deployments.

NUMBER 2

95%

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

The 7-stage ML lifecycle.

2

DevOps vs MLOps — 5 differences.

3

CI/CD pipeline for a model repo.

4

5 artefacts every MLOps system versions.

5

Configure GitHub Actions + MLflow.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The 7-stage ML lifecycle

Problem → Data → Features →
Model → Deploy → Monitor →
Feedback

02

DevOps vs MLOps

Key concept of this lesson.

03

5 artefacts that must be...

Code, data, features, models,
experiments

04

CI/CD for ML — the pipeline

Gated at every stage

01

PART 1 OF 4

The 7-stage ML lifecycle

Problem → Data → Features → Model → Deploy → Monitor → Feedback

CONCEPT 1 OF 4

The 7-stage ML lifecycle

Problem: define + success metric.

Data: collect, clean, label, version.

Features: engineer, document, store.

Model: train, validate, register.

Deploy: package, serve, canary.

Monitor: drift, performance, cost.

Feedback: retrain triggers; labels from prod.

LOOP

7 stages · closes back

80% of effort is stages 2-3 + 5-7, not training.

DEFINITION

The 7-stage ML lifecycle

Problem → Data → Features → Model → Deploy →
Monitor → Feedback

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

DevOps vs MLOps

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
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5 artefacts that must be versioned

Code — Git (as always).

Data — DVC, LakeFS.

Features — feature store registries (Feast, Tecton).

Models — MLflow, W&B, SageMaker Registry.

Experiments — every run with hyperparams + metrics.

MEMORISE

5 artefacts

Skip any one = unreproducible.

DEFINITION

5 artefacts that must be versioned

Code, data, features, models, experiments

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

CI/CD for ML — the pipeline

1. Push code → lint + unit tests.
2. Data validation (Great Expectations).
3. Train; log metrics to MLflow.
4. Evaluate vs current prod; gate merge.
5. Build + push image.
6. Canary deploy 5%.
7. Auto-promote if SLOs hold 30 min.

GATE

Nothing auto-promotes

Without passing SLOs. Ever.

KEY POINTS

- 1. Push code → lint + unit tests
- 2. Data validation (Great Expectations)
- 3. Train; log metrics to MLflow
- 4. Evaluate vs current prod; gate merge

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Correct order — 7-stage lifecycle starts with:



Model



Data



Problem



Deploy

Correct order — 7-stage lifecycle starts with:



CORRECT ANSWER

Problem

WHY

Problem definition + success metric drives everything

Typical DevOps artefact vs MLOps:

A

Same

B

Code only vs 5 artefacts

C

Docker only

D

Different cloud

Typical DevOps artefact vs MLOps:

CORRECT ANSWER



Code only vs 5 artefacts

WHY

MLOps versions code + data + features + models + experiments

CI gate before merge:

A

Lint only

B

New model F1 $\geq$ current prod - 0.5%

C

Any change OK

D

Random

CI gate before merge:

CORRECT ANSWER



New model F1 $\geq$ current prod - 0.5%

WHY

Prevent regressions; gate on model-metric threshold

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Problem: define + success metric.

Data: collect, clean, label, version.

Features: engineer, document, store.

Model: train, validate, register.

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LOOP

7 stages • closes back

80% of effort is stages 2-3 + 5-7, not training.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1

The 7-stage ML lifecycle

2

DevOps vs MLOps

3

5 artefacts that must be...

4

CI/CD for ML — the pipeline

Train the model. Then version everything that made it.

"The 95% is the job."

DELIVERABLES

- NEXT
- Lesson 2 — Docker + Kubernetes.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.